Patuakhali Science and Technology University
Faculty of Computer Science and Engineering

CIT 324 :: Simulation & Modeling Sessional

Project Proposal

Project Title : Advanced Epidemiological Disease-Spread Simulation
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Submitted from,	Submitted to,
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Contents

EpiSim

1. Introduction

The integration of mathematical modeling into public health is a critical component of modern epidemiological planning. These models provide the foundational framework required to predict infectious disease trajectories, estimate healthcare resource demands, and measure the effectiveness of interventions like vaccination and social distancing. However, the accessibility of these predictive tools is frequently hindered by high computational barriers. Executing complex models typically requires advanced programming proficiency in languages such as Python or R, creating a significant disconnect between the data scientists who build the models and the public health officials, medical researchers, and students who actually need to use them.

To address this gap, we present EpiSim, an interactive, academic-grade disease spread simulation platform. Built utilizing Python and Streamlit, EpiSim is engineered to democratize mathematical epidemiology by providing a strictly no-code, visually driven analytical ecosystem. It seamlessly translates theoretical compartmental models—specifically the SIR, SIS, SEIR, and SEIRD architectures—into dynamic, animated visual trajectories that map how individuals transition between health states over time.

2. Objectives.

(a) To build an interactive, no-code Streamlit dashboard for configuring and visualizing compartmental mathematical models (SEIRD, SEIR, SIR, SIS) in real-time.

(b) To implement a robust SciPy ODE integration engine that accurately computes disease trajectories, time-varying reproduction numbers (R_t), and herd immunity thresholds.

(c) To enable empirical data calibration by utilizing numerical optimizers (L-BFGS-B / Nelder-Mead) to automatically extract best-fit parameters from user-uploaded outbreak datasets.

(d) To generate publication-quality scientific visualizations—including S-I phase planes and sensitivity tornados—and provide seamless CSV data export for academic reporting.

3. Problem Statement

Modeling disease outbreaks requires analyzing both complex mathematical trajectories and the practical impact of public health interventions. Existing tools fail to bridge this gap: advanced computational libraries solve complex ODEs accurately but demand steep coding skills, while accessible web calculators lack the scientific depth required for empirical data fitting or advanced SEIRD dynamics. As a result, critical public health questions—such as estimating ICU demand under social distancing—are either gatekept by software engineers or answered using overly simplistic models. There is a critical need for a unified, no-code platform that visually simulates complex epidemic trajectories and policy outcomes side-by-side, delivering academic-grade robustness in a highly accessible interface.

4. Related Work

The development of EpiSim is grounded in nearly a century of mathematical epidemiology, originating from the foundational compartmental SIR models introduced by Kermack and McKendrick [1] and expanded into the comprehensive SEIRD dynamics detailed by Anderson and May [2]. To accurately capture varying transmission behaviors, the platform's computational engine balances deterministic population-level differential equations with the stochastic simulation frameworks established by Gillespie [3] and emphasized by Keeling and Rohani [4]. Furthermore, EpiSim's analytical modules for evaluating Non-Pharmaceutical Interventions (NPIs) and empirically tracking the time-varying reproduction number, R_t, are directly informed by the critical COVID-19 modeling methodologies developed by Ferguson et al. [5] and Flaxman et al. [6]. By unifying these established theoretical and stochastic frameworks into a streamlined, interactive application, EpiSim successfully transitions complex mathematical research into an accessible, real-time tool for data-driven public health planning.

5. Methodology

5.1. Technology Stack

Development followed an iterative, agile approach. The computational stack was selected to maximize numerical precision, ensure interactive real-time responsiveness, and support seamless mathematical modeling without requiring a code execution layer on the user end.

Component	**Technology**
Frontend Dashboard Framework	Streamlit (Python-native reactive UI)
Mathematical Engine	SciPy (ndimage & integrate modules)
Numerical Optimization Backend	SciPy Optimize (L-BFGS-B and Nelder-Mead routines)
Interactive Visualization Layer	Plotly Python (Dynamic SVG/WebGL rendering)
Data Manipulation & Parsing	Pandas + NumPy
Deterministic Solvers	Runge-Kutta 4th/5th Order (RK45 ODE Solver)
Stochastic Solvers	Gillespie Direct Algorithm Engine
Architecture & Report Layout	Mermaid.js; report typeset in Typst

5.2. Design Principles

(a) Decoupled mathematical engine: All epidemiological equations (ODEs), stochastic solvers, and optimization logic (L-BFGS-B) live in independent, purely mathematical Python modules.

(b) Reactive parameter state: A centralized Streamlit session state acts as the single source of truth for all epidemiological variables (Population, $b\eta$, Σ, γ, μ). Any adjustment to a slider instantly propagates through the SciPy solver and triggers a synchronized, real-time update across all five analytical tabs.

(c) Local, offline-first execution: The platform requires no external databases, user accounts, or cloud API calls.

(d) Mathematically bounded inputs: To prevent impossible epidemiological states, all interactive inputs are strictly typed and bounded.

6. System Modeling and Visual Architecture

6.1. Flow Chart Diagram

Figure 1: Flow Chart of NeuroVision AI Architecture

The Figure 1 flowchart diagram illustrates the illustrates the high-level user interaction and data processing pathways within the EpiSim.

6.2. Project Timeline(Gantt Chart)

Task	Wk 1	Wk 2	Wk 3	Wk 4	Wk 5	Wk 6
Mathematical models + SEIRD ODEs + SciPy tests	✓					
Streamlit state store + parameter integration		✓				
Dashboard UI, Plotly charts, and metric cards		✓	✓			
NPI comparisons, Phase Space, R_t tracking			✓	✓		
L-BFGS-B optimization & empirical data fitting					✓	
CSV exports, testing, and final typesetting					✓	✓

6.3. User Interface

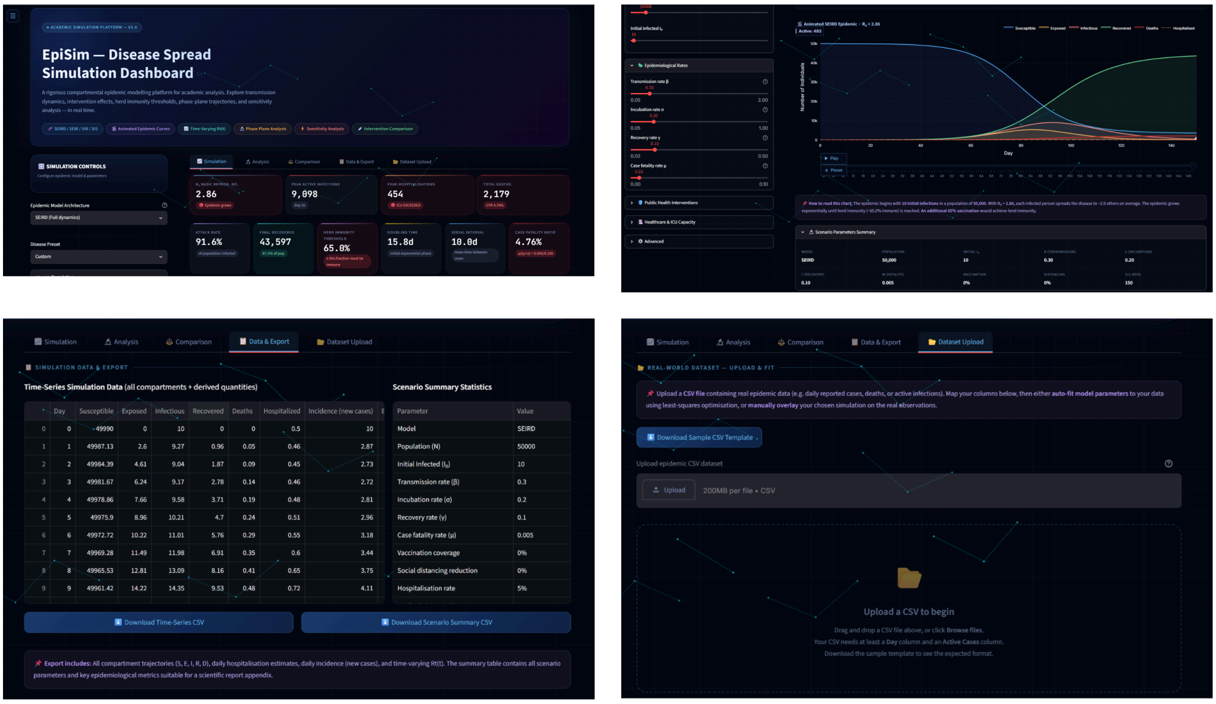

Figure 2: EpiSim User Interface modules.

7. Future Plans

While EpiSim successfully delivers a robust, localized foundation for mathematical epidemiology, future development iterations will focus on expanding its analytical complexity and predictive capabilities. Key planned enhancements include:

1. **Spatial and Network Dynamics:** Transitioning from strictly well-mixed population models to multi-node network architectures. This will allow users to simulate disease spread across distinct geographic regions, interconnected by transit routes or commuter data.

2. **Multi-Strain Pathogen Modeling:** Extending the core SEIRD differential equations to support concurrent, competing viral variants, enabling the analysis of varying transmission rates and vaccine-evasion profiles within a single outbreak.
3. **Machine Learning Integration:** Augmenting the traditional SciPy optimization backend with deep learning techniques (such as Physics-Informed Neural Networks) to enhance long-term epidemic forecasting when empirical surveillance datasets are sparse or noisy.
4. **Cloud-Based Collaboration:** Building an optional, secure cloud-hosted deployment tier that allows multiple public health officials to collaboratively edit simulation parameters, share interactive dashboards, and compile joint reports in real-time.

8. Result

EpiSim successfully translates complex epidemiological mathematics into an accessible, real-time simulation platform powered by SciPy ODE solvers. By automating advanced empirical data fitting and generating dynamic visualizations—such as phase portraits and time-varying reproduction number (R_t) charts—the system enables researchers and policymakers to seamlessly calibrate outbreak models and immediately quantify the impact of targeted public health interventions.

9. References

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[2] R. M. Anderson and R. M. May, *Infectious Diseases of Humans: Dynamics and Control*. Oxford University Press, 1991.

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[5] N. Ferguson and others, "Report 9: Impact of non-pharmaceutical interventions (NPIs) to reduce COVID-19 mortality and healthcare demand," technical report, 2020.

[6] S. Flaxman and others, "Estimating the effects of non-pharmaceutical interventions on COVID-19 in Europe," *Nature*, vol. 584, pp. 257–261, 2020.